

SCPN-MIF-CORE: an open, formally verified deterministic trigger and interlock lane for pulsed field-reversed-configuration magneto-inertial fusion

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Summary

SCPN-MIF-CORE is an open, machine-checked reference implementation of an independent deterministic trigger and interlock lane for pulsed field-reversed-configuration (FRC) magneto-inertial fusion. It occupies the layer beneath machine-learning and model-predictive controllers: prescribed kinematic and diagnostic inputs are reduced to a merge-window decision and a synthesisable SystemVerilog trigger fabric. Safety, liveness, and bounded-cycle timing properties are checked with k-induction in the open Yosys and SymbiYosys flow (Wolf and YosysHQ 2024), while Verilator cosimulation compares the software and RTL paths. The package is deliberately neither a plasma-physics solver nor a plasma controller; it consumes those systems’ outputs through explicit interfaces.

The verified RTL surface includes absolute kinematic-safety veto dominance, single-shot behaviour, debounce no-underflow, zero-cycle veto dominance for a registerless fast-veto lane, bounded lock-to-trigger cycle latency, and a two-flop clock-domain-crossing synchroniser property at the sensor-stream ingress. Proof-status manifests fail on drift so that continuous-integration success requires the configured proofs to run again.

Statement of need

FRC programmes increasingly combine physics models, diagnostics, and learned or optimising control components (Belova et al. 2025). Those components benefit from a separate deterministic layer that can permit or inhibit a trigger under an absolute safety envelope. Low-latency FPGA inference for physics triggering is an active field; for example, the cited `hls4m1` radiation-hard FPGA study

reports a synthesis-indicated latency of 25 ns on a PolarFire target (Govorkova et al. 2026). Commercial pulsed-fusion programmes also develop solid-state switching and trigger hardware (Moffett and Chesny 2026). SCPN-MIF-CORE contributes an open trigger/interlock implementation with reproducible register-transfer-level properties and software-to-RTL parity checks.

The package does not claim lower latency than specialised inference frameworks or equivalence to an operating fusion device. Reported latency evidence is cycle-accurate and modelled unless a result explicitly states otherwise; it is not post-route silicon timing. The repository also reproduces the published merge/no-merge kinematic classification of an FRC merging-and-compression study (Belova et al. 2025), while delegating reconnection-driven merge-time physics to its solver dependency instead of fitting that phenomenon inside the trigger lane.

Functionality and verification

The Python API and command-line interface include a deterministic demonstration scenario that produces a decision without an external input file. Numerical kernels provide a Rust acceleration path with parity checks, and selected discrete invariants are represented in Lean 4. The synthesisable SystemVerilog lane has golden-reference and Verilator cosimulation tests. The formal lane uses Yosys and SymbiYosys, with named properties and non-vacuity witnesses recorded in a proof-status manifest.

Continuous integration exercises the public software surface, formal tasks, bit-true cosimulation, documentation, dependency locks, and security checks. Compatibility and evidence manifests are validated against their schemas to reduce silent contract drift between the trigger lane and sibling components.

Availability and evidence boundary

Source code, tests, proof harnesses, reproducibility commands, and the review materials are available in the public repository. The manuscript describes software, model, proof, synthesis, and local RTL-cosimulation evidence. It does not report a post-route implementation, calibrated data-acquisition campaign, hardware-in-the-loop campaign, facility deployment, plasma gain, or produced energy. Those require separately identified hardware, calibration, custody, and experimental evidence.

This manuscript is a review draft. Its presence in the repository does not mean that it has been submitted to, accepted by, or published in a journal or preprint service.

Acknowledgements

Timing-aware and clock-domain-crossing formal-property templates are maintained by the sibling **sc-neurocore** project and reused under the shared SCPN ecosystem.

References

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